Power

File: power.kicad_sch

Level 2 drive

File: level2.kicad_sch

Head

File: head.kicad_sch

Piper-Watch wiring. Power: the 19 V feed, split block, panel fuse, switch and buck.

Level 2 drive: the perfboard (TMC2209), the display's ESP32-S3, Hall sensor and motor. Head: the XIAO and the three displays.

Nets that cross sheets use global labels: +19V, +19V_FUSED, +19V_SW, +5V, GND. Everything else is local to its sheet.

USB (data + logic power) runs to the Jetson from: the JC4827 display, head XIAO ESP32-S3, C920 camera board.

Wiring draft generated from hardware/wiring.md

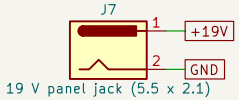
Piper-Watch

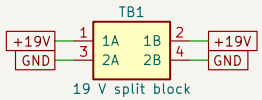
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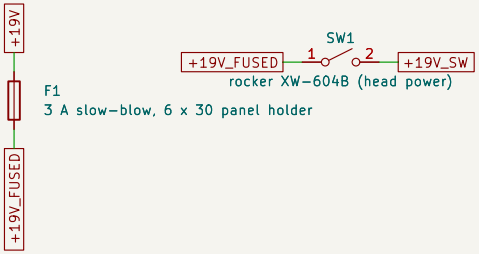
File: piper-watch.kicad_sch

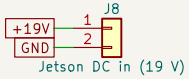
Title: Piper-Watch wiring

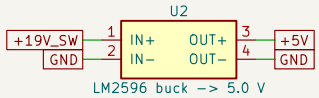
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/4











19 V supply -> panel jack -> split block TB1 (on Level 2's floor). TB1's − side is the COMMON GROUND.

The Jetson is fed straight from TB1: always on, never switched or fused here (the supply limits it).

Head side: TB1 + -> panel fuse F1 (6 x 30 mm, 3 A slow-blow) -> rocker switch SW1 -> perfboard J1 (+19V_SW).

Fuse holder: TB1 to its END contact, its SIDE contact to the switch (an unscrewed cap's exposed end is dead).

U2 (LM2596 buck, on its own floor bosses) is fed from perfboard J6; set it to 5.0 V before connecting the Hall sensor.

Wiring draft generated from hardware/wiring.md

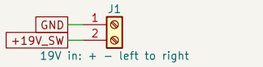
Piper-Watch

Sheet: /Power/

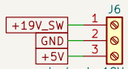
File: power.kicad_sch

Title: Power: 19 V in, split block, panel fuse, switch, 5 V buck

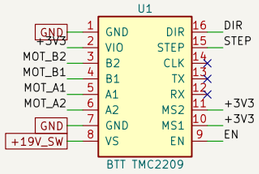
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/4



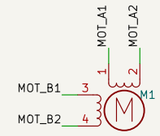
19V in: + − left to right



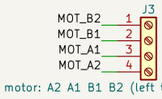
to buck: 19V, GND, 5V



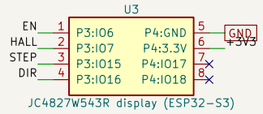
BTT TMC2209



NEMA17 (42 x 33.4)



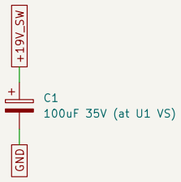
motor: A2 A1 B1 B2 (left to right on the board)

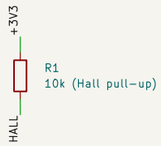


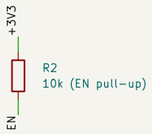
JC4827W543R display (ESP32-S3)

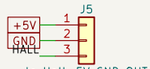


to display P3/P4: STEP DIR EN HALL 3V3 GND

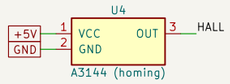








to Hall: 5V GND OUT



A3144 (homing)

PERFBOARD (70 x 50): screw terminals J1 (19 V in), J3 (motor), J6 (buck); pin headers J4 (display P3/P4), J5 (Hall) with Dupont housings.

Heavy wire (20–22 AWG): 19 V, GND, motor. C1 sits right at U1's VS and the GND next to it.

U1 MS1 = MS2 = 3V3: 1/16 microstepping. CLK, TX, RX and DIAG unconnected. Set Vref ~0.7 V before connecting the motor.

R2 holds EN high (motor off) while the display's ESP32 boots — fit it.

Motor M1: 6-pin socket, pins 1+3 = coil A (red A1, blue A2); pins 4+6 = coil B (green B1, black B2); pins 2, 5 (centre taps) unused. J3 left to right: A2 A1 B1 B2.

U3 (the JC4827W543R display's ESP32-S3) runs the motor: P3 + P4 pigtails -> J4. Powered by its USB from the Jetson — never also from the buck.

Hall U4 (A3144) on the motor bracket's pad: VCC from the 5 V buck, OUT pulled up to 3V3 by R1 (never to 5 V).

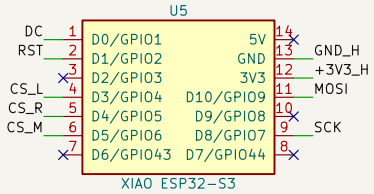
Wiring draft generated from hardware/wiring.md

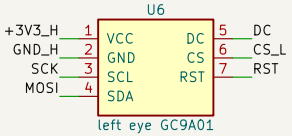
Piper-Watch

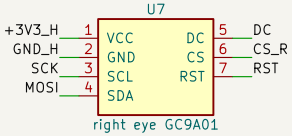
Sheet: /Level 2 drive/
File: level2.kicad_sch

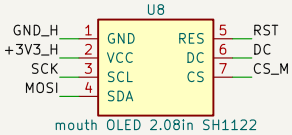
Title: Level 2: perfboard (TMC2209), the display's ESP32-S3, Hall sensor, m

Size: A3	Date:	**Rev:**
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The XIAO and the C920 camera each have their own USB cable to the Jetson, through the turntable's hollow.

No Wi-Fi/BT: the XIAO's antenna stays unplugged. Displays run from the XIAO's 3V3 (~150–250 mA total).

One SPI bus (SCL, SDA, DC, RST shared); a CS per display. 26–28 AWG silicone wire soldered straight to the pads.

The eyes mount upside down: the firmware rotates their image 180 deg. No backlight pin on these modules. D2 (GPIO3, strapping), D6, D7, D9 free.

Wiring draft generated from hardware/wiring.md

Piper-Watch

Sheet: /Head/

File: head.kicad_sch

Title: Head: XIAO ESP32-S3, two GC9A01 eyes, SH1122 mouth

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/4